

Practical Task (L3T02)

My Database Theory

1. **Data** refers to raw, unprocessed facts and figures that have no meaning.
Information is data that has been processed, structured and given context so that it has a meaning on it.
2. **Metadata** is often described as data about data. It provides descriptive context that helps make raw data understandable and usable. For example, a dataset containing temperature readings such 10, 15, 20 and 25 becomes meaningful when metadata tells us these are day temperatures measured in °C for the week.
3. A **Database Management System (DBMS)** is a collection of software programs designed to manage database structures and control access to the data stored within them. It serves as a layer of facilitating between the end user and the database itself, translating user requests into the internal processes needed to access or modify data.

Advantages of DBMS:

- The DBMS controls who can access what data, enforcing privacy and security policies while still allowing efficient data retrieval for authorized users.
- Centralized data management gives users a consistent and unified view of organizational data.
- The DBMS enables quick responses to requests for reading, inserting, updating, and deleting data.
- Reliable and well-managed data.

4. **Operational Databases (OLTP):**

- Designed to support the day-to-day running of an organization.
- They store current, detailed and frequently updated information
- Optimized for speed and high-volume transactions
- An example is an inventory management system at a retail store, which must track stock levels in real time as sales and deliveries occur.

Analytical Databases (OLAP):

- They are built for analysis and business intelligence rather than live transactions.
- They store historical and aggregated data.
- Optimized for complex queries and reporting
- An example is a data warehouse used by a retail chain to analyze sales trends over several years.

5. **Types of Data:**

- **Unstructured data:** Text documents, photos, videos and social media posts that do not fit neatly into rows and columns.
- **Semi-structured data:** Data in formats like JSON or XML that has some organization but lacks the rigid structure of relational tables.
- **Large-scale distributed data:** Data sets that need to scale horizontally across many servers, such as those generated by high-traffic web applications.

Common cases where NoSQL databases are most effective:

- Real-time web applications that require rapid reading and writing at massive scale
- Content management systems storing diverse media types

- Internet of Things (IoT) applications generating continuous streams of sensor data.
 - E-commerce platforms that need flexible product catalogues where items have different features.
6. **SQLite** is the DBMS that does not require server configuration. It is a lightweight, file-based relational database that operates in a serverless mode and stores the entire database as a single file on disk.

Advantages of using SQLite:

- There is no server to install, configure and maintain which makes setup extremely simple.
 - The entire database is a single file that can be copied, moved and shared easily across systems.
 - It offers solid performance for applications that do not require concurrent multi-user access.
 - It requires no external dependencies which reduces complexity in development environments.
7. **ACID** is an acronym which stands for **Atomicity, Consistency, Isolation, Durability**.

In the context of a DBMS,

- **Atomicity:** A transaction is treated as a single and indivisible unit. Either all operations within the transaction are completed successfully or none of them are applied.
- **Consistency:** Every transaction brings the database from one valid state to another. The rules and constraints of the database must always be maintained ensuring the data remains correct and reliable.
- **Isolation:** Transactions that run concurrently are executed in a way that does not allow them to interfere with each other. The intermediate state of

a transaction is not visible to other transactions which prevent data corruption in multiple user transactions.

- **Durability:** Once a transaction has been done, its changes are permanently saved even if the system crashes immediately afterward. This is achieved through mechanisms like transaction logs and backups.